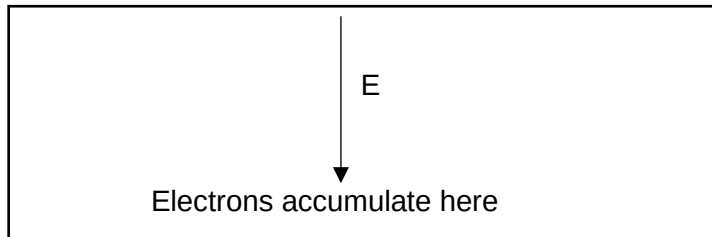


4(a) Magnetic force $F_B = Bev = (4.0 \times 10^{-3}) \times (1.60 \times 10^{-19}) \times (0.60 \times 10^{-3}) = 3.84 \times 10^{-25} \text{ N}$

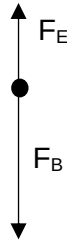
(b)



[Note: Direction of conventional current is opposite to that of electron flow.
Using FLHR, magnetic force is downwards, so electrons accumulate at the bottom.]

(c)(i) See above diagram.

(ii) 1.



- The electric force F_E points upwards [APPLICATION].
- As the electric field gets stronger, F_E also gets stronger [APPLICATION]. The net force $F_{\text{net}} = F_E - F_B$ gets weaker [CONCEPT: net force = vector addition]. So the rate of accumulation drops [CONCLUSION].
- Eventually $F_E = F_B$, there's no more net force acting on the electron [APPLICATION], and the accumulation stops [CONCLUSION].

2. Accumulation stops: $eE = Bev$ [CONCEPT]
 $E = Bv$

Potential Difference, $V = Ed = Bvd$ [CONCEPT]
 $= (4.0 \times 10^{-3}) \times (0.60 \times 10^{-3}) \times (1.5 \times 10^{-2}) = 3.6 \times 10^{-8} \text{ V}$

5(a) Higher frequency/lower wavelength means higher energy photons **OR $E = hf$**